PLUMBING ABBREVIATIONS

A	
ABBR	ABBREVIATION
ADA	AMERICANS WITH DISABILITIES ACT
AFF	ABOVE FINISHED FLOOR
AG	ABOVE GRADE
AHJ	AUTHORITY HAVING JURISDICTION
AMB	AMBIENT
ANSI	AMERICAN NATIONAL STANDARDS INSTITUTE
AP	ACCESS PANEL
APPROX	APPROXIMATE
B	
BFP	BACKFLOW PREVENTOR (TYPE AS INDICATED)
BFV	BUTTERFLY VALVE
BHP	BRAKE HORSEPOWER
BLV	BALANCING VALVE
BTUH	BRITISH THERMAL UNIT PER HOUR
BV	BALL VALVE
C	
CA	UTILITY COMPRESSED AIR (PRESSURE AS INDICATED)
CAP	CAPACITY
CFM	CUBIC FEET PER MINUTE
CI	CAST IRON
CO	CLEANOUT
COMP	COMPRESSOR
COND	CONDENSATE
CONT	CONTINUATION
CV	CHECK VALVE
CW	DOMESTIC COLD WATER
D	
DF	DRINKING FOUNTAIN
DIA	DIAMETER
DN	DOWN
DOM	DOMESTIC
DWG	DRAWING
DWH	DOMESTIC WATER HEATER
DWR	DRINKING WATER RETURN
DWS	DRINKING WATER SUPPLY
E	
(E)	EXISTING MATERIAL OR EQUIPMENT
EA	EACH
ECO	EXTERIOR CLEANOUT
EEW	EMERGENCY EYE WASH
EFF	EFFICIENCY
EFW	EMERGENCY FACE WASH
EL	ELEVATION
EQ	EQUIVALENT
ESH	EMERGENCY SHOWER
ET	EXPANSION TANK
EWC	ELECTRIC WATER COOLER
EWT	ENTERING WATER TEMPERATURE
EXP	EXPANSION
EXT	EXTERNAL
F	
F	DEGREES FAHRENHEIT
FCO	FLOOR CLEANOUT
FD	FLOOR DRAIN
FL	FLOOR
FPM	FEET PER MINUTE
FPS	FEET PER SECOND
FS	FLOOR SINK
FT	FEET OR FOOT
FTD	FOOTER DRAINAGE
G	
G	NATURAL GAS (PRESSURE AS INDICATED)
GAL	GALLON
GC	GAS COCK
GPD	GALLONS PER DAY
GPH	GALLONS PER HOUR
GPM	GALLONS PER MINUTE
GV	GATE VALVE
GW	GREASE WASTE
H	
HB	HOSE BIBB
HG	MERCURY
HGT	HEIGHT
HP	HORSE POWER
HPS	HIGH PRESSURE STEAM
HR	HOUR
HW	DOMESTIC HOT WATER
HWR	DOMESTIC HOT WATER RETURN
I	
ID	INSIDE DIAMETER
IE	INVERT ELEVATION
IN.	INCHES
K	
kW	KILOWATTS
L	
LAV	LAVATORY
LGT	LENGTH
LPG	LIQUID PETROLEUM GAS
LPS	LOW PRESSURE STEAM
LWT	LEAVING WATER TEMPERATURE
M	
MAX	MAXIMUM
MBH	1000 BTU/HR
MCA	MINIMUM CURRENT AMPACITY
MIN	MINIMUM
MISC	MISCELLANEOUS
MOCP	MINIMUM OVER CURRENT PROTECTION
MPS	MEDIUM PRESSURE STEAM
MTG	MOUNTING
N	
NA	NOT APPLICABLE
NC	NORMALLY CLOSED
NIC	NOT IN CONTRACT
NO	NORMALLY OPEN/ NUMBER
NOM	NOMINAL
NTS	NOT TO SCALE

O	
OD	OUTSIDE DIAMETER
OD	OVERFLOW ROOF DRAIN
OFL	OVERFLOW
OSD	OPEN SITE DRAIN
P	
P	PUMP
PC	PUMPED CONDENSATE
PD	PUMPED DISCHARGE (TYPE AS INDICATED)
PDI	PLUMBING DRAINAGE INSTITUTE
PH	PHASE (ELECTRICAL)
PPM	PARTS PER MILLION
PRESS	PRESSURE
PRV	PRESSURE REDUCING VALVE
PSI	POUNDS PER SQUARE INCH
PSIG	POUNDS PER SQUARE INCH - GAUGE
PV	PLUG VALVE
PW	PROCESS WATER
R	
RD	ROOF DRAIN
REF	REFERENCE
RL	RELOCATED MATERIAL OR EQUIPMENT
RPM	REVOLUTIONS PER MINUTE
RV	PRESSURE RELIEF VALVE
S	
SAN	SANITARY OR WASTE
SCFM	STANDARD CUBIC FEET PER MINUTE
SCW	SOFTENED COLD WATER
SE	SEWAGE EJECTOR
SG	SPECIFIC GRAVITY
SP	SUMP PUMP
SPEC	SPECIFICATION
SQ. FT.	SQUARE FEET
SS	STAINLESS STEEL
SSD	SUB-SOIL DRAINAGE
ST	STORM (RAINWATER)
STD	STANDARD
STM	STEAM
SV	SOLENOID VALVE
T	
TEMP	TEMPERATURE
TP	TRAP PRIMER
TPU	TRAP PRIMER UNIT
TPV	TEMPERATURE AND PRESSURE RELIEF VALVE
TW	TEMPERED WATER
TYP	TYPICAL
U	
UON	UNLESS OTHERWISE NOTED
V	
V	VOLTAGE/VENT
VAC	VACUUM
VEL	VELOCITY
VTR	VENT THRU ROOF
W	
W	WITH
WO	WITHOUT
WC	WATER CLOSET
WCO	WALL CLEAN OUT
WG	WATER GAUGE
WH	WALL HYDRANT
WHA	WATER HAMMER ARRESTOR
Y	
YD	YARD

SEE PLUMBING SYMBOLS, LEGENDS AND EQUIPMENT SCHEDULES FOR ADDITIONAL ABBREVIATIONS. ALL ABBREVIATIONS, SYMBOLS, AND LEGENDS SHOWN ON THIS DRAWINGS ARE NOT NECESSARILY USED.

PLUMBING FLOOR PLAN SYMBOLS

ISOLATION VALVE	REDUCER - CONCENTRIC
BALANCING VALVE	REDUCER - ECCENTRIC
ANGLE VALVE	PIPE UNION
CHECK VALVE	PIPE ANCHOR
SOLENOID VALVE	PIPE ALIGNMENT GUIDE/ PIPE SLEEVE
BUTTERFLY VALVE	
BALL VALVE	PRESSURE GAUGE WITH COCK
PLUG VALVE or GAS COCK	THERMOMETER
PRESSURE REGULATOR VALVE FLANGED	QUICK COUPLING
PRESSURE REGULATOR VALVE THREADED	CAP OR PLUG
FLOW INDICATING BALANCE VALVE	PIPE UP
PRESSURE RELIEF VALVE or TEMP PRESSURE RELIEF VALVE	PIPE DOWN
STRAINER	TEE OUTLET - DOWN
MECHANICAL EXPANSION JOINT	BRANCH TOP CONNECTION
FLEXIBLE CONNECTION	DIRECTION OF FLOW
CLEANOUT	FLOW METER
FLOOR CLEANOUT or EXTERIOR CLEANOUT	AIR VENT WITH VALVE
FLOOR DRAIN (ROUND)	WATER HAMMER ARRESTOR
FS FLOOR SINK (RECTANGULAR)	HOSE BIBB
ROOF DRAIN	WALL HYDRANT
MATERIAL OR EQUIPMENT TO BE DEMOLISHED	POINT OF CONNECTION
MATERIAL OR EQUIPMENT EXISTING TO REMAIN	POINT OF DISCONNECT

PLUMBING FLOOR PLAN SYMBOLS

DCW DOMESTIC COLD WATER	V SANITARY VENT
DHW DOMESTIC HOT WATER	SAN SANITARY DRAIN
DHWR DOMESTIC HOT WATER RETURN	SAN UNDER SLAB SANITARY DRAIN

REFERENCE SYMBOLS

1 / E0.1 — SECTION NUMBER; DRAWING NUMBER WHERE DRAWN

DETAIL NUMBER; DRAWING NUMBER WHERE DRAWN

MXXX — KEY NOTE

HHWS / 4 — RISER DESIGNATION (TYPE AS INDICATED)

4 — DEMO KEY NOTE

PLUMBING GENERAL NOTES

1. INSTALL PLUMBING IN ACCORDANCE WITH THE INTERNATIONAL PLUMBING CODE 2018 AND REGULATIONS OF ALL AUTHORITIES HAVING JURISDICTION.
2. OBTAIN COMPLETE CONSTRUCTION DOCUMENTS WHICH INCLUDE BUT ARE NOT LIMITED TO ARCHITECTURAL, STRUCTURAL, MECHANICAL, PLUMBING, ELECTRICAL, AND FIRE PROTECTION DRAWINGS. ALSO OBTAIN A COMPLETE SET OF SPECIFICATIONS.
3. DURING THE COORDINATION PROCESS BRING TO THE ATTENTION OF THE ARCHITECT AND ENGINEER(S) ANY CONFLICTS. THE CONTRACTOR SHALL NOT PROCEED WITH ANY INSTALLATION EXCEPT AT HIS OWN RISK UNTIL ALL CONFLICTS ARE RESOLVED AND THE CLARIFYING INFORMATION ISSUED TO THE CONTRACTOR BY THE ARCHITECT.
4. THE CONTRACTOR SHALL VERIFY AND COORDINATE THE LOCATION OF ALL STRUCTURAL MEMBERS FOR CONFLICTS WITH PIPING. THE CONTRACTOR SHALL NOTIFY THE ARCHITECT IMMEDIATELY IF A CONFLICT IS DETERMINED.
5. PROVIDE CLEANOUTS WHERE INDICATED ON DRAWINGS AND WHERE REQUIRED BY THE APPLICABLE CODES.
6. FIRE STOP ALL OPENINGS IN COMPLIANCE WITH THE U.L. LISTING OF THE METHOD USED TO MAINTAIN THE APPROPRIATE FIRE RATING AS REQUIRED BY CODE.
7. COORDINATE ALL PIPING INSTALLED WITH ALL OTHER TRADES (I.E. STRUCTURAL, MECHANICAL, ELECTRICAL, AND FIRE PROTECTION) TO ENSURE THAT ALL SYSTEMS ARE INSTALLED ABOVE THE CEILING OR IN A CONCEALED LOCATION. ALL CEILING HEIGHTS ARE INDICATED ON THE ARCHITECTURAL AND/OR INTERIOR DESIGN DRAWINGS. MAINTAIN MINIMUM CLEARANCES REQUIRED BY THE APPLICABLE CODES THROUGHOUT THE BUILDING.
8. PROVIDE TRAP PRIMERS AND DISTRIBUTION UNITS OR A MECHANICAL ALTERNATIVE I.E. TRAP GUARD AS INDICATED ON DRAWINGS. INSTALL DISTRIBUTION UNITS LEVEL AND LOCATED IN AN ACCESSIBLE AND PROTECTED LOCATION.
9. PROVIDE WATER HAMMER ARRESTORS WHERE QUICK CLOSING VALVES ARE UTILIZED. LOCATE ARRESTORS PER MANUFACTURER'S INSTRUCTIONS. ARRESTOR SHALL BE ACCESSIBLE.
10. INSULATE ALL STORM DRAIN BODIES AND HORIZONTAL PIPING. THE HORIZONTAL STORM PIPING SHALL BE INSULATED AT THE UPPER MOST LEVEL.
11. INSULATE ALL DRAINAGE PIPING RECEIVING CONDENSATE.
12. INSULATE ALL DRAINAGE AND WATER PIPING IN COMPLIANCE WITH ANSI 117.1, LOCAL CODES AND SPECIFICATIONS.
13. INSTALL VALVES AND OR ACCESSORIES SERVING EQUIPMENT TO ALLOW FOR THE REMOVAL OR SERIVICING OF THE VALVES,ACCESORIES AND EQUIPMENT.
14. WHEN PROVIDING UNIONS, VACUUM BREAKERS, STRAINERS, ETC. AS INDICATED ON DRAWINGS AND ALL EQUIPMENT, PROVIDE FOR MAINTENANCE CLEARANCE AS REQUIRED BY CODE.
15. REFER TO THE ARCHITECTURAL DRAWINGS FOR EXACT LOCATIONS OF ALL PLUMBING FIXTURES.
16. SEE THE PLUMBING FIXTURE CONNECTION SCHEDULE FOR CONNECTION SIZES.
17. ALL ABOVE CEILING WATER/LIQUID 1/4 TURN VALVES SHALL BE HORIZONTAL WITH THEIR VALVE HANDLE. PULL DOWN TOWARDS OPERATOR FOR SHUTOFF.

DEMOLITION GENERAL NOTES

1. THE CONTRACTOR SHALL VERIFY ALL EXISTING CONDITIONS DURING THE CONTRACTOR'S PRE-BID SITE VISIT(S) AND BRING TO THE ATTENTION OF THE ARCHITECT/ENGINEER ANY DISCREPANCIES BETWEEN THE CONTRACT DOCUMENTS AND THE ACTUAL SITE CONDITIONS PRIOR TO BID FOR CLARIFICATION VIA PRE-BID RFI.
2. THE REQUIRED DEMOLITION IS NOT LIMITED TO THAT PORTION INDICATED ON THE PLANS ALONE, BUT SHALL INCLUDE ALL NECESSARY WORK INDICATED ELSEWHERE IN THE SPECIFICATIONS WHICH ARE NECESSARY TO COMPLETE THE INTENT OF THE CONTRACT DOCUMENTS.
3. THE CONTRACTOR SHALL NOT VIOLATE THE PHYSICAL SECURITY OF THE BUILDING DURING DEMOLITION OR ASSOCIATED OPERATIONS. COORDINATE DEMOLITION WITH OWNER/AGENT.
4. THE CONTRACTOR SHALL SCHEDULE WITH THE OWNER/AGENT ANY INTERRUPTIONS OF EXISITNG UTILITIES. PRIOR TO STARTING WORK THE CONTRACTOR SHALL HAVE OBTAINED WRITTEN PERMISION TO PROCEED. THE CONTRACTOR SHALL ASSUME THE BUILDING IS OCCUPIED AND OPERATIONAL AT ALL TIMES.
5. THE CONTRACTOR SHALL COORDINATE WITH ALL OTHER TRADES.
6. IT IS UNDERSTOOD THAT WORK UNDER THESE CONTRACT DOCUMENTS MAY EXTEND PAST THE LIMITS OF WORK. THE OWNER SHALL BE NOTIFIED PRIOR TO WORK STARTING IN THESE AREAS AND PROVIDE A SCHEDULE.
7. ALL FIXTURES AND EQUIPMENT INDICATED TO BE REMOVED SHALL HAVE ALL ASSOCIATED PIPING REMOVED BACK TO THE NEAREST BRANCH, MAIN, RISER AND CAPPED IN A CONCEALED LOCATION.
8. "CONCEALED LOCATION" IS DEFINED AS BEING ABOVE, BELOW OR BEHIND A FINISHED PLANE.
9. ALL FIXTURES, EQUIPMENT, PIPING AND APPURTANCES SHALL BECOME THE PROPERTY OF THE CONTRACTOR AND SHALL BE REMOVED IN A TIMELY MANNER. THE SALVAGE VALUE SHALL APPEAR AS A CREDIT IN THE BID.
10. ANY PIPE SHALL BE CAPPED USING A COMMERCIALLY PURCHASED CAP. A VALVE OR CRIMPING SHALL NOT CONSTITUTE A CAP AND IS UNACCEPTABLE.

PIPE MATERIAL SCHEDULE

SANITARY WASTE/VENT PIPE (ABOVE GRADE):	• CISPI SERVICE WEIGHT CAST-IRON, NO-HUB AND FITTINGS WITH NEOPRENE GASKETS AND STAINLESS-STEEL BANDS
SANITARY WASTE/VENT PIPE (BELOW GRADE):	• CISPI SERVICE WEIGHT CAST-IRON, NO-HUB AND FITTINGS WITH NEOPRENE GASKETS AND STAINLESS-STEEL BANDS • PVC SCHEDULE 40 SOLID WALL AND PVC DWV FITING SYSTEMS AND SOLVENT WELD JOINTS.
DOMESTIC WATER PIPE (ABOVE GRADE):	• ASTM B88 AND ANSI/NSF 61 TYPE 'L' HARD DRAWN COPPER PIPE AND FITTINGS WITH LEAD FREE SOLDER JOINTS. (INSULATE DOMESTIC HOT WATER AND CIRCULATION PIPING).
DOMESTIC WATER PIPE (BELOW GRADE):	• SOFT COPPER NO JOINTS
CONDENSATE DRAIN PIPE:	• TYPE 'M' HARD DRAWN, COPPER PIPE AND FITTINGS WITH LEAD FREE SOLDER JOINTS (INSULATE PIPE WITHIN THE BUILDING).

UCCS

UCCS CYBERSECURITY AND SPACE ECOSYSTEM EXPANSION

3650 North Nevada Ave
Colorado Springs, CO, 80907

SMITHGROUP

899 LOGAN STREET
SUITE 508
DENVER, CO 80203
303.832.3272
smithgroup.com

RIDER LEVETT BUCKNALL
COST ESTIMATOR
1675 LARIMER STREET
DENVER, CO, 80202
720-904-1480

NV5
ACOUSTICS, IT/TELECOM & SECURITY, AUDIOVISUAL
2650 18TH STREET, SUITES 201-203
DENVER, CO 80211
720-213-7550

MARTIN/MARTIN, INC.
STRUCTURAL ENGINEER
12499 WEST COLFAX AVENUE
LAKEWOOD, CO, 80215
303-431-6100

ISSUED FOR	REV	DATE
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DESIGN DEVELOPMENT		23DEC2020
SCHEMATIC DESIGN		19OCT2020

SEALS AND SIGNATURES

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